Arithmetic Rank and Differential Complexity of Hamiltonian Moment Integrals

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Abstract

For the diagonal Hamiltonian circle action on a product of n symplectic two-spheres, we determine the scalar differential complexity of the exponential moment integral after all collisions of fixed-point values have been combined. If the moment lengths have rational rank r, the minimal differential order is at least $2^{r-1}(n-r+2)$. This bound is attained for every $1 \le r \le n$. The combinatorial input is a sharp support bound for a product of n Laurent binomials spanning rank r, proved by counting vertices and edge-interior coefficients of its Newton zonotope. We also give an exact formula when the rational relation space is one-dimensional. For the uncentered Hamiltonian moments, we classify every polynomial-coefficient recurrence: its order is at least one less than the surviving exponential support, and each coefficient has a specified degree-n divisor. The differential Galois group is a torus of dimension r. These results distinguish the number of fixed points, the number of critical levels, and the number of analytically visible levels. The localization and spline formulas are classical; the paper supplies complete proofs of the support and complexity statements and does not claim a resolution of a named open conjecture.

Keywords. Hamiltonian circle action; Duistermaat–Heckman measure; zonotope; sparse polynomial; rational rank; differential equation; polynomial recurrence.

1 The problem and the main result

The exponential integral of a Hamiltonian moment map is constrained by the fixed points of the action. What is less immediate is how many of those constraints survive when several fixed points have the same moment value. An upper differential equation obtained by multiplying all fixed-point factors can substantially overestimate the minimum order. We investigate that loss of complexity for products of symplectic spheres.

Let $a = (a_1, \dots, a_n)$ with $a_i > 0$, and define

$$F_a(t) = \prod_{i=1}^{n} \frac{e^{a_i t} - 1}{a_i t}, \qquad F_a(0) = 1. \tag{1.1}$$

Section 2 realizes this function as a normalized symplectic integral. For a subset $S \subseteq [n]$, write $a(S) = \sum_{i \in S} a_i$. Set

$$\begin{aligned} c_\lambda &= \sum_{S:\, a(S)=\lambda} (-1)^{n-|S|}, \qquad \Sigma_a = \{\lambda : c_\lambda \neq 0\}, \\ N(a) &= |\Sigma_a|, \qquad\qquad r(a) = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{a_1, \dots, a_n\}. \end{aligned} \tag{1.2}$$

Thus 0 and $A = \sum_i a_i$ belong to Σ_a. All its other elements are positive and smaller than A. Define $\operatorname{dord} F$ as the least order of a nonzero scalar linear differential operator over $\mathbb{C}(t)$ annihilating F.

Theorem 1.1 (Arithmetic rank and differential complexity)**.** *For every positive tuple a,*

$$\operatorname{dord} F_a = N(a) \geq 2^{r(a)-1}(n - r(a) + 2). \tag{1.3}$$

For every pair $1 \leq r \leq n$, positive lengths of rational rank r attain equality. In particular, rationally independent lengths give order 2^n, and order $n+1$ forces all lengths to be commensurable when $n \geq 2$.

Writing $F_a(t) = \sum_{k\geq 0} m_k t^k/k!$, the minimum order of a polynomial-coefficient recurrence for (m_k) is $N(a) - 1$, and the minimum coefficient degree among all such recurrences is n. The differential Galois group of the minimal differential equation over $\mathbb{C}(t)$ is an algebraic torus of dimension $r(a)$.

Here commensurable means that every ratio a_i/a_j is rational. The converse to the order-$n+1$ assertion is false: for example, lengths $(1, 2)$ are commensurable but yield four exponential terms. The recurrence statement uses the normalization $\min H = 0$; shifting a Hamiltonian preserves differential order but can change recurrence order.

1.1 Proof dependencies and relation to earlier work

The analytic transfer used here is specific. The grouped local-connection formula for weighted Jacobi pushforwards in [8, Part IX, theorem "Weighted graphical Jacobi local connections"] specializes to the present setting by taking every endpoint exponent and interaction exponent to be zero, the analytic amplitude to be one, and the linear weights to be a_i. Equal subset sums must then be combined with their signs. The dual-singular-support lemma of [9, Section 3.1] turns each surviving non-smooth level into a lower bound on differential order. Its negative-exponent transfer principle motivates the recurrence divisibility statement below. We reproduce these specializations and their proofs; no unverified classification from either manuscript is needed as a black box.

The additional ingredient is geometric: a Newton zonotope and its central hyperplane arrangement force enough coefficients to survive, even when subset sums collide. It is this step that produces the rational-rank bound, which does not follow merely by counting fixed points. The exact corank-one formula separately identifies which single rational relations produce cancellation.

Duistermaat–Heckman localization and its equivariant formulation are classical [1, 2]. In the product model, they reduce to integration over intervals; the resulting density is a univariate box spline [3]. The binomial products themselves belong to the established literature on pure products, root systems, and discrete tomography [5, 6, 7]. We do not present those connections, the spline formula, or exponential independence as new. A focused literature comparison did not establish historical priority for the combined sharp rank-and-complexity statement. This manuscript records a self-contained theorem and proof, not a certified priority or publication-level assessment.

2 The Hamiltonian model and visible critical levels

Equip each sphere $S_i^2 \simeq \mathbb{CP}^1$ with a rotationally invariant symplectic form and the standard rotation action. Normalize the angular coordinate to have period one and the moment coordinate x_i to range over $[0, a_i]$. On the complement of the poles, the symplectic form is $dx_i \wedge d\theta_i$. Consequently its pushforward under x_i is Lebesgue measure on $[0, a_i]$. On

$$M_a = (S^2)^n, \qquad \omega_a = \sum_i \pi_i^* \omega_i,$$

the diagonal circle action has moment map $H_a = \sum_i x_i$ and 2^n isolated fixed points. Its normalized Liouville measure pushes forward to the convolution of the uniform probability measures on $[0, a_i]$. This proves (1.1) directly. The endpoint lengths may be arbitrary positive real numbers; an integral symplectic class is not assumed.

Proposition 2.1 (Exact collision and visibility formula)**.** *Put $V = \prod_i a_i$. The pushforward density satisfies, as a distribution,*

$$\rho_a(x) = \frac{1}{V(n-1)!} \sum_{S \subseteq [n]} (-1)^{|S|} (x - a(S))_+^{n-1}, \tag{2.1}$$

with $x_+^0 = \mathbf{1}_{x>0}$ when $n = 1$. In particular,

$$D_x^n \rho_a = \frac{(-1)^n}{V} \sum_{\lambda \in \Sigma_a} c_\lambda \delta_\lambda. \tag{2.2}$$

The density is polynomial on each component of $\mathbb{R} \setminus \Sigma_a$. At each $\lambda \in \Sigma_a$, its $(n-1)$st derivative has a nonzero jump. At a subset sum with $c_\lambda = 0$, it is polynomial across that level. Thus Σ_a is exactly its non-smooth locus.

Proof. For one interval, $D\mathbf{1}_{[0,a_i]} = \delta_0 - \delta_{a_i}$. Differentiate the convolution once in each factor to obtain $D^n \rho_a = V^{-1} *_{i=1}^n (\delta_0 - \delta_{a_i})$. The distribution $(x-b)_+^{n-1}/(n-1)!$ is the unique nth primitive of δ_b vanishing on the far left, giving (2.1). Grouping equal subset sums gives (2.2). The local polynomial and jump assertions follow immediately. □

Lemma 2.2 (Dual singular support)**.** *Let ρ be a compactly supported distribution and $F(t) = \langle \rho, e^{tx} \rangle$. If $L = \sum_{j=0}^q p_j(t) \partial_t^j \neq 0$ has polynomial coefficients and $LF = 0$, set $d = \max_j \deg p_j$ and $\chi_L(x) = \sum_j [t^d] p_j(t) x^j$. Then ρ is real analytic away from the zeros of the nonzero polynomial χ_L, whose degree is at most q.*

Proof. Fourier–Laplace injectivity gives $\widehat{L}\rho = 0$, where $\widehat{L} = \sum_j p_j(-D_x) \circ x^j$. Its highest derivative coefficient is $(-1)^d \chi_L(x)$. If $d = 0$, this says $\chi_L \rho = 0$. If $d > 0$, away from its zeros the distributional equation is an ordinary analytic first-order system for $(\rho, D\rho, \dots, D^{d-1}\rho)$. Multiplying by the inverse of an analytic fundamental matrix shows that this vector is an analytic matrix times a constant vector. Hence ρ is analytic there. □

Proposition 2.3 (Minimum scalar order)**.** *Let $P_a(z) = \prod_{\lambda \in \Sigma_a} (z - \lambda)$. The monic minimal operator of F_a over $\mathbb{C}(t)$ is*

$$L_a = t^{-n} P_a(\partial_t)\, t^n. \tag{2.3}$$

It has order $N(a)$ and no finite singularity outside zero.

Proof. Since $V t^n F_a = \sum_\lambda c_\lambda e^{\lambda t}$, the displayed operator annihilates F_a and has order $N(a)$. Conversely, Proposition 2.1 and Lemma 2.2 force the top-degree polynomial of every polynomial annihilator to vanish at all $N(a)$ visible levels. Its order is at least $N(a)$. Clearing rational denominators does not change order. The displayed monic operator must therefore be minimal; its coefficients have poles only at zero. □

The lower bound just proved is the actual use of the earlier visibility method. Formula (2.1) is its collision-sensitive input. Counting the 2^n fixed points without grouping would give a false answer, as the examples below show.

3 A sharp support bound for products of binomial differences

The following argument is independent of symplectic localization. All coefficients are taken in a field of characteristic zero.

Theorem 3.1. *Let $v_1, \dots, v_n \in \mathbb{Z}^r \setminus \{0\}$ span $\mathbb{R}^r$, and put*

$$B(X) = \prod_{i=1}^{n} (1 - X^{v_i}) \in \mathbb{C}[X_1^{\pm 1}, \dots, X_r^{\pm 1}].$$

Then

$$|\operatorname{supp} B| \geq 2^{r-1}(n - r + 2).$$

This bound is attained for every $1 \leq r \leq n$.

Proof. Partition the vectors into k parallel classes, of respective sizes $m_1, \dots, m_k$, so that $\sum_j m_j = n$. The Newton polytope of B is the full-dimensional zonotope

$$Z = \sum_{i=1}^{n} [0, v_i].$$

The identity follows either from the usual product rule for Newton polytopes or directly by exposing a vertex with a generic linear functional: in each binomial the exposed endpoint is unique, so the coefficient of the exposed vertex cannot cancel.

We first count vertices. This is the standard central-arrangement minimum-region argument; see also the extremal-arrangement literature [4]. Let H_j be the hyperplane orthogonal to the jth parallel class. The vertices of Z correspond to the chambers of the central arrangement $\{H_1, \dots, H_k\}$. Start with r hyperplanes whose normals are independent. These have 2^r chambers. Adding a further distinct hyperplane H increases the number of chambers by the number of chambers of the arrangement induced on H. The induced arrangement is essential in H: the restrictions to H of the original r independent normal forms span H^*. It therefore has at least 2^{r-1} chambers. Consequently the number V of vertices satisfies

$$V \geq 2^r + (k - r)2^{r-1} = 2^{r-1}(k - r + 2).$$

For $r = 1$ there is just one parallel class and $V = 2$, consistent with the same formula.

For each class j, a functional in H_j avoiding all the other hyperplanes exposes an edge parallel to that class. The resulting edges are indexed by the chambers of the induced arrangement on H_j. Thus the number E_j of these edges is at least 2^{r-1}. In dimension one, take the segment Z itself as its unique edge, so that $E_1 = 1$.

On any such edge, the face polynomial of B is a Laurent monomial times the product of precisely the m_j binomials from the jth class. Writing the class vectors as nonzero integer multiples of a primitive lattice vector identifies this product, up to a Laurent monomial, with

$$\prod_{i=1}^{m_j} (1 - z^{q_i}), \qquad q_i \in \mathbb{Z} \setminus \{0\}.$$

It has a zero of exact order m_j at $z = 1$. A nonzero Laurent polynomial with s nonzero terms has vanishing order at most $s - 1$ there: applying $(z\partial_z)^a$ for $a = 0, \dots, s - 1$ gives an invertible Vandermonde matrix in its distinct exponents. The face polynomial therefore has at least $m_j + 1$ terms. Its two extreme terms are the edge endpoints, leaving at least $m_j - 1$ support points in the relative interior of the edge.

Distinct edges have disjoint relative interiors. Counting vertices and these edge-interior support points gives

$$\begin{aligned}|\operatorname{supp} B| &\geq V + \sum_{j=1}^{k} E_j(m_j - 1) \\ &\geq 2^{r-1}(k - r + 2) + 2^{r-1}(n - k) = 2^{r-1}(n - r + 2).\end{aligned}$$

Finally, choose $n - r + 1$ copies of e_1, and one copy of each of $e_2, \ldots, e_r$. The resulting polynomial is $(1 - X_1)^{n-r+1} \prod_{j=2}^{r}(1 - X_j)$, whose support has exactly $(n - r + 2)2^{r-1}$ elements. □

Corollary 3.2. *Let $a_1, \ldots, a_n > 0$, let $r = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{a_1, \ldots, a_n\}$, and combine equal exponents in $\prod_i (e^{a_i t} - 1) = \sum_{s \in \Sigma} c_s e^{st}$, with every $c_s \neq 0$. Then*

$$|\Sigma| \geq 2^{r-1}(n - r + 2).$$

In particular, $|\Sigma| = n + 1$ forces $r = 1$ when $n \geq 2$.

Proof. Choose a rational basis for the lengths and clear all denominators to write $a_i = \langle v_i, b \rangle$ with $v_i \in \mathbb{Z}^r$ and the coordinates of b linearly independent over $\mathbb{Q}$. Distinct integer exponent vectors then give distinct real exponents, so the preceding theorem applies without loss of support. For $r \geq 2$, its bound is at least $2n > n + 1$; indeed $2^{r-1}(n - r + 2) \geq 2n$ for $n \geq r \geq 2$. □

Proposition 3.3 (Three positive lengths)**.** *For three positive lengths, $|\Sigma| = 4$ if and only if, up to permutation and common scale, their multiset is $(1, 1, 1)$ or $(1, 1, 2)$.*

Proof. Write $a \leq b \leq c$ and $A = a+b+c$. If $a < b$, the coefficients at a and b in $\prod(e^{a_i t} - 1)$ are both nonzero: no pair of the three lengths can sum to either value. Their reflected exponents $A - a$ and $A - b$ are also nonzero, by antisymmetry about $A/2$. These four exponents are distinct and lie strictly between 0 and A, yielding at least six support points. Thus $a = b$ is necessary. For (a, a, c) the candidate exponents are $0, a, 2a, c, c + a, c + 2a$. If $c > a$ and $c \neq 2a$ they are all distinct, so there are six terms. If $c = a$, expansion gives four terms, and if $c = 2a$ the terms at $2a$ cancel, again leaving four. □

4 Exact collisions in rational corank one

The sharp support bound admits a complete refinement when there is only one rational relation among the parameters. This refinement distinguishes three phenomena: a rational dependence can produce no subset-sum collision; a collision can merge two coefficients without cancellation; or it can cancel both coefficients.

For a tuple $a = (a_1, \ldots, a_n)$ of positive real numbers, write

$$G_a(t) = \prod_{i=1}^{n}(e^{a_i t} - 1) = \sum_{\lambda} c_\lambda e^{\lambda t}, \qquad N(a) = \#\{\lambda : c_\lambda \neq 0\},$$

where equal exponents have been collected before $N(a)$ is counted. The integral relation lattice is

$$\mathcal{R}(a) = \left\{ u \in \mathbb{Z}^n : \sum_{i=1}^{n} u_i a_i = 0 \right\}.$$

If $\dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{a_i\} = n - 1$, then $\mathcal{R}(a) = \mathbb{Z}u$ for a primitive integer vector u, unique up to sign. Here primitive means that the greatest common divisor of its coordinates is one.

Theorem 4.1 (Exact support in rational corank one)**.** *Let $n \geq 2$ and let $a_1, \ldots, a_n > 0$ have rational rank $n-1$. Let u be a primitive generator of $\mathcal{R}(a)$.*

1. *If $|u_i| > 1$ for some i, then every subset sum of the a_i is distinct and*
$$N(a) = 2^n.$$
2. *Otherwise, every nonzero coordinate of u belongs to $\{-1, 1\}$. Put $c = \#\{i : u_i \neq 0\}$. Then $c \geq 2$ and*
$$N(a) = \begin{cases} 2^n - 2^{n-c}, & c \text{ even}, \\ 2^n - 2^{n-c+1}, & c \text{ odd}. \end{cases} \tag{4.1}$$

Consequently,
$$N(a) \geq 3 \cdot 2^{n-2}, \tag{4.2}$$
with equality if and only if, after permuting coordinates and changing the sign of u, the unique rational relation is of one of the forms
$$a_1 = a_2, \qquad \text{or} \qquad a_1 + a_2 = a_3.$$
The second possibility requires $n \geq 3$. In both cases the statement that the relation is unique is part of the hypothesis: all remaining parameters are independent modulo that relation.

Proof. For a subset $S \subseteq \{1, \ldots, n\}$ let $\mathbf{1}_S \in \{0,1\}^n$ be its indicator vector and put $a(S) = \sum_{i \in S} a_i$. If $a(S) = a(T)$, then
$$\mathbf{1}_S - \mathbf{1}_T \in \mathcal{R}(a) = \mathbb{Z}u.$$
For completeness, the multiplier is an integer, not merely rational: if $qu \in \mathbb{Z}^n$ and u is primitive, a Bezout combination of the coordinates of u shows that $q \in \mathbb{Z}$.

Every coordinate of $\mathbf{1}_S - \mathbf{1}_T$ lies in $\{-1, 0, 1\}$. If some $|u_i| > 1$, no nonzero multiple of u can have this property. Thus $S = T$ in every collision, proving the first assertion.

Assume now that the nonzero coordinates of u are all ± 1, and set
$$A = \{i : u_i = 1\}, \qquad B = \{i : u_i = -1\}, \qquad C = \{1, \ldots, n\} \setminus (A \cup B).$$
The positivity of the a_i ensures that A and B are both nonempty. The relation is $a(A) = a(B)$ and $c = |A| + |B| \geq 2$.

If two distinct subsets have equal sums, their indicator difference is u or $-u$: larger integer multiples are impossible inside the binary cube. After exchanging the subsets if necessary, they must therefore be
$$A \cup T \quad \text{and} \quad B \cup T, \qquad T \subseteq C. \tag{4.3}$$
Conversely, every pair in (4.3) has equal sums. There are exactly 2^{n-c} such pairs. Their common sums are distinct from one another and from all other subset sums. Indeed, every fiber of the subset-sum map has at most two elements: three distinct elements on an affine line with direction u would include two whose difference has a coordinate of absolute value at least two.

The contribution of a subset S to G_a is $(-1)^{n-|S|} e^{a(S)t}$. For the two subsets in (4.3), the ratio of the signs is $(-1)^{|A|-|B|} = (-1)^c$. When c is even, the two equal exponents merge into one term with a nonzero coefficient. Each pair therefore decreases the support count by one. When c is odd, the coefficients cancel and each pair decreases the support count by two. This proves (4.1).

If c is even, then $c \geq 2$ and the deficit from 2^n is at most 2^{n-2}, with equality precisely when $c = 2$. If c is odd, then $c \geq 3$ and the deficit is again at most 2^{n-2}, with equality precisely when $c = 3$. Positivity turns the corresponding primitive relations into $a_i = a_j$ and $a_i + a_j = a_k$, respectively. The case with a coordinate $|u_i| > 1$ has support 2^n and is never extremal. □

Example 4.2. Let $a, b > 0$ be rationally independent. The tuple $(a, b, a+b)$ has six surviving exponents: its one collision cancels the two terms at $a+b$. The tuple $(a, b, 2a+b)$ has the same rational rank but all eight subset sums are distinct. Rational dependence alone therefore does not imply a loss of exponential support.

For three positive rationally independent numbers a, b, c, the tuple $(a, b, c, a+b)$ attains the corank-one minimum 12. In contrast, $(a, b, c, a+b+c)$ has support 15: its unique colliding pair has equal signs and merges rather than cancels.

5 All polynomial recurrences and their unavoidable factors

We first establish the scalar statement needed for the geometric application. Write $D = d/dt$ and

$$h_n(k) = (k+1)(k+2)\cdots(k+n).$$

A polynomial recurrence has the form $\sum_{j=0}^{M} p_j(k) m_{k+j} = 0$ for all sufficiently large integers k, with $p_0 p_M \neq 0$. Its order is M and its coefficient degree is $\max_j \deg p_j$.

Theorem 5.1 (Exponential numerator and all polynomial recurrences)**.** *Let $n \geq 1$, let $0 = \lambda_0 < \lambda_1 < \cdots < \lambda_s$, and let all $c_0, \ldots, c_s$ be nonzero real numbers. Suppose*

$$G(t) = \sum_{i=0}^{s} c_i e^{\lambda_i t}$$

has a zero of exact order n at 0. Put $F(t) = t^{-n} G(t)$ and $m_k = F^{(k)}(0)$. Define

$$P(z) = \prod_{i=0}^{s} (z - \lambda_i), \qquad Q(z) = \prod_{i=1}^{s} (z - \lambda_i) = \sum_{j=0}^{s} q_j z^j.$$

Then the following assertions hold.

1. *$s \geq n$. The minimal order over $\mathbb{C}(t)$ of a nonzero linear differential operator annihilating F is $s+1$. Its monic minimal operator is*

   $$L = t^{-n} \circ P(D) \circ t^n.$$

   The polynomial-coefficient operator $L_{\mathrm{pol}} = P(D) \circ t^n$ is primitive, has leading coefficient t^n, and has no finite singularity except possibly 0.

2. *The indicial roots of L at 0 are*

   $$-n, -n+1, \ldots, s-n.$$

   Writing $\theta = tD$, the operator $t^{s+1-n} L_{\mathrm{pol}}$ has θ-span s, with lowest θ-coefficient $(\theta+n)^{\underline{s+1}}$.

3. *The moments satisfy*

   $$\sum_{j=0}^{s} q_j h_n(k+j) m_{k+j} = 0 \qquad (k \geq 0).$$

   This recurrence has minimal order s and minimal coefficient degree n among all eventual polynomial-coefficient recurrences.

4. *More precisely, for arbitrary $M \geq 0$ and $p_0, \ldots, p_M \in \mathbb{C}[k]$, the identity*

   $$\sum_{j=0}^{M} p_j(k) m_{k+j} = 0$$

holds for all sufficiently large integers k if and only if

$$h_n(k+j) \mid p_j(k) \quad (0 \le j \le M), \qquad Q(z) \mid \sum_{j=0}^{M} \frac{p_j(k)}{h_n(k+j)} z^j \quad \text{in } \mathbb{C}[k,z].$$

Consequently every recurrence of order s is a polynomial in k times the displayed recurrence, and its degree-n representative is unique up to a nonzero constant.

Proof. The $s+1$ by $s+1$ Vandermonde matrix in the distinct λ_i shows that a nonzero sum of these exponentials cannot have a zero of order greater than s. Hence $s \ge n$.

Distinct real exponentials are linearly independent over $\mathbb{C}(t)$. Indeed, in any nontrivial relation, a nonzero rational coefficient of the largest exponential behaves as a nonzero constant times a power of t as $t \to +\infty$, whereas all terms of smaller exponential rate decay exponentially after division by that largest exponential. This is impossible. If an operator $A = \sum_{j=0}^{m} a_j(t) D^j$ annihilates G, this independence implies $\sum_j a_j(t) \lambda_i^j = 0$ for every i. When $m < s+1$, a polynomial of degree at most m cannot vanish at all $s+1$ distinct λ_i, unless all a_j vanish. Thus $P(D)$ is the monic minimal operator of G, and rational gauge conjugation proves the assertion for F.

The leading coefficient of $P(D) \circ t^n$ is t^n. The coefficient of D^{s+1-n} has nonzero constant term $\binom{s+1}{n} n!$, arising from the leading term of $P(D)$; lower-degree terms of P contribute only positive powers of t to that coefficient. Thus no factor of t divides every coefficient, and primitivity follows. If $P(z) = \sum_{j=0}^{s+1} b_j z^j$, then

$$t^{s+1-n} L_{\mathrm{pol}} = \sum_{j=0}^{s+1} b_j t^{s+1-j} (\theta+n)^{\underline{j}}.$$

Here $b_0 = 0$, $b_1 \neq 0$, and $b_{s+1} = 1$. This proves the span and indicial assertions. It also proves regular singularity at 0.

Coefficient extraction gives

$$m_k = \frac{k!}{(k+n)!} \sum_{i=0}^{s} c_i \lambda_i^{k+n} = \frac{1}{h_n(k)} \sum_{i=1}^{s} d_i \lambda_i^k, \qquad d_i = c_i \lambda_i^n \neq 0.$$

The zero phase disappears because $k+n > 0$. Consequently $b_k = h_n(k) m_k$ is annihilated by the constant-coefficient shift operator $Q(E)$, where $(Eb)_k = b_{k+1}$. This gives the claimed recurrence.

It remains to prove the classification, including its saturation at every negative integer pole. Suppose an eventual recurrence holds. The sequences λ_i^k are linearly independent over $\mathbb{C}(k)$, by the same largest-rate argument along positive integers. Therefore, for every $i = 1, \dots, s$,

$$R_i(k) := \sum_{j=0}^{M} \frac{p_j(k)}{h_n(k+j)} \lambda_i^j = 0 \quad \text{in } \mathbb{C}(k).$$

Fix an integer ℓ. The terms with a possible pole at $k = -\ell$ are precisely the shifts

$$j \in J_\ell = [\ell - n, \ell - 1] \cap [0, M] \cap \mathbb{Z}.$$

All these poles are simple and J_ℓ contains at most n consecutive integers. Taking residues in the identities $R_i = 0$ gives a polynomial in z of the form

$$A_\ell(z) = \sum_{j \in J_\ell} r_j z^j$$

which vanishes at all s nonzero numbers λ_i. After removing its smallest possible power of z, its degree is at most $n-1<s$. It is therefore identically zero. Every residue r_j vanishes. For a fixed shift j, varying $\ell = j+1,\dots,j+n$ proves that p_j vanishes at every root of $h_n(k+j)$, so that $h_n(k+j) \mid p_j(k)$.

The resulting polynomial $A(k,z) = \sum_j [p_j(k)/h_n(k+j)]z^j$ vanishes at every λ_i. Division by the monic constant-coefficient polynomial $Q(z)$ in $\mathbb{C}[k][z]$ proves $Q \mid A$. Conversely this divisibility annihilates every exponential component and yields the recurrence. It forces $M \geq s$ for any nonzero recurrence with nonzero extreme coefficients. It also forces every nonzero p_j to have degree at least n. If $M = s$, its quotient by Q is a polynomial depending only on k, proving the uniqueness assertions. □

Indicial transfer made explicit. For the symplectic function, take $G = Vt^nF_a$ and divide the moments in the theorem by V. The negative indicial roots are precisely $-1,\dots,-n$. Together with the absence of nonzero finite singularities, they are the input to the indicial-to-recurrence transfer method of [9, Section 2]. The residue argument above reproves that mechanism here and strengthens it to a divisor for every shift coefficient, not merely an order or degree bound.

6 The differential Galois torus

The order $N(a)$ counts surviving characters. The rational rank counts how many independent multiplicative parameters those characters require. These two quantities are different even when every subset sum is distinct.

Proposition 6.1. *The Picard–Vessiot field of L_a over $\mathbb{C}(t)$ is*

$$K_a = \mathbb{C}(t)(e^{\lambda t} : \lambda \in \Sigma_a) = \mathbb{C}(t)(e^{a_1 t},\dots,e^{a_n t}),$$

and its differential Galois group is isomorphic to $(\mathbb{C}^\times)^{r(a)}$. In particular $\operatorname{trdeg}_{\mathbb{C}(t)} K_a = r(a)$.

Proof. A fundamental set of solutions is $\{t^{-n}e^{\lambda t} : \lambda \in \Sigma_a\}$. Differentiating them does not enlarge the displayed field, and the reciprocal Wronskian belongs to it. Its constants are $\mathbb{C}$, since its elements are meromorphic functions with their usual derivative.

In fact the integer span of Σ_a equals the integer span of the a_i. The first inclusion follows from the subset-sum expansion. For the reverse inclusion, induct on the distinct lengths b in increasing order. If the coefficient at exponent b survives, then $b \in \Sigma_a$. Otherwise the nonzero sum of singleton contributions at b must be cancelled by a subset using only strictly smaller lengths. Such a subset expresses b in the integer span already obtained by induction. This proves the integer-span identity and the second field equality.

The additive subgroup generated by Σ_a is free abelian of rank $r = r(a)$. Choose its integer basis $b_1,\dots,b_r$. Then $K_a = \mathbb{C}(t)(e^{b_1 t},\dots,e^{b_r t})$: both directions follow by writing generators as integer combinations. The b_j are rationally independent. A polynomial relation among these exponentials over $\mathbb{C}(t)$ would give a rational linear relation among finitely many distinct real exponentials. Such a relation is impossible by the largest-rate argument. The generators are therefore algebraically independent. A differential automorphism must send each $e^{b_j t}$ to a nonzero constant times itself, and every independent choice of these constants gives an automorphism of this rational function field. This proves the assertion. □

This is an elementary diagonal differential module, not a claim of a new general Galois-theoretic mechanism. Together with the support bound, it gives the constraint

$$\operatorname{dord} F_a \geq 2^{\dim \operatorname{Gal}(L_a)-1}(n - \dim \operatorname{Gal}(L_a) + 2)$$

within this Hamiltonian family. This finishes the proof of Theorem 1.1, once the recurrence theorem of the preceding section is applied to Vt^nF_a.

7 Examples, invisible fixed-point levels, and extremizers

7.1 A critical level that disappears analytically

For $a = (1, 1, 2)$, expansion gives

$$(e^t - 1)^2(e^{2t} - 1) = e^{4t} - 2e^{3t} + 2e^t - 1. \tag{7.1}$$

There are eight fixed points and five distinct critical values $0, 1, 2, 3, 4$, but only four visible values. At level 2, the subset $\{3\}$ and the subset $\{1, 2\}$ have opposite signs and cancel. The normalized density is

$$\rho(x) = \frac{1}{4}[x_+^2 - 2(x-1)_+^2 + 2(x-3)_+^2 - (x-4)_+^2].$$

It is the single quadratic $(-x^2 + 4x - 2)/4$ throughout $(1, 3)$. Thus level 2 is genuinely absent from the non-smooth locus.

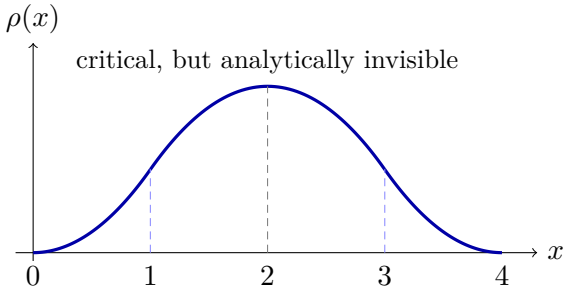


Figure 1: The pushforward density for lengths $(1, 1, 2)$. The derivative of order two jumps at $0, 1, 3, 4$, and has no jump at 2.

Here $P(z) = z(z-1)(z-3)(z-4)$, so the minimal differential order is four. With $h_3(k) = (k+1)(k+2)(k+3)$, the minimal moment recurrence is

$$\begin{aligned} &-12h_3(k)m_k + 19h_3(k+1)m_{k+1} \\ &\qquad - 8h_3(k+2)m_{k+2} + h_3(k+3)m_{k+3} = 0. \end{aligned}$$

The initial values $m_0 = 1$, $m_1 = 2$, $m_2 = 9/2$ follow from the sum of three uniform variables. For instance, the recurrence at $k = 0$ gives $m_3 = 11$.

7.2 Complete order classification for three spheres

For three positive lengths, only the orders $4, 6, 8$ occur:

$$N(a, b, c) = \begin{cases} 4, & (a, b, c) \text{ is proportional, up to permutation, to } (1, 1, 1) \text{ or } (1, 1, 2), \\ 6, & \text{otherwise, two lengths are equal or one is the sum of the other two,} \\ 8, & \text{otherwise.} \end{cases} \tag{7.2}$$

Indeed, positivity means that a collision of distinct subsets can only come from equal singleton lengths or from a singleton equalling a disjoint pair. If neither event occurs, all eight sums are distinct. Exactly two equal lengths, without the exceptional ratios above, leave six nonzero terms. A relation $a + b = c$ with $a \neq b$ cancels exactly the two terms at c, also leaving six. The exceptional cases were proved in Proposition 3.3.

7.3 Sharpness for every dimension and rational rank

Choose positive, rationally independent numbers $b_1, \dots, b_r$ and take the tuple

$$a = (\underbrace{b_1, \dots, b_1}_{n-r+1 \text{ copies}}, b_2, \dots, b_r).$$

Its numerator is $(e^{b_1 t} - 1)^{n-r+1} \prod_{j=2}^{r} (e^{b_j t} - 1)$, and no exponents from different lattice points collide. Therefore $N(a) = (n - r + 2)2^{r-1}$, proving sharpness in the positive symplectic setting, not merely for abstract Laurent polynomials.

For the fixed manifold $(S^2)^n$, rationally independent lengths give $N = 2^n = \chi((S^2)^n)$, whereas equal lengths give $N = n + 1$. Hence differential order is not a topological invariant of the underlying manifold. It reflects the arithmetic of the symplectic parameters and the cancellations in the chosen moment map.

7.4 Root-system examples and a classical obstruction

Repeated lengths are not the only way to attain the sharp bound. Let $a, b > 0$ be rationally independent. The following tuples have rational rank two and attain $N = 2n$:

$$(a, b), \quad (a, b, a + b), \quad (a, b, a + b, a + 2b),$$

and

$$(a, b, a + b, 2a + b, 3a + b, 3a + 2b).$$

Their lattice vectors are the positive-root patterns of $A_1 \times A_1$, A_2, B_2, and G_2. The alternating Weyl denominator identities, or direct multiplication, give respectively $4, 6, 8, 12$ surviving monomials. These classical root-system cancellations are already part of the pure-product literature [6]; their role here is to furnish extremal Hamiltonian examples.

There is a useful limitation on this list, inherited from discrete tomography. Suppose $r = 2$ and the n integer exponent vectors are pairwise nonparallel. If the support has size $2n$, it consists exactly of the $2n$ vertices of the Newton polygon. Every coefficient is then 1 or -1, and the positive and negative supports each have n points. Divisibility by each binomial implies that sums of the signed coefficients on every line in that direction vanish. Thus the two supports have the same discrete X-rays in all n directions. Alpers–Larman's lattice theorem [7, Theorem 2.2] implies that equality is impossible for $n = 5$ or $n > 6$. Consequently the possible values are exactly $n \in \{2, 3, 4, 6\}$, with existence supplied by the displayed examples. This is a consequence of their theorem, not a new classification of lattice switching sets or of all extremizing configurations.

8 Scope and further questions

The symplectic application is to products of spheres with the diagonal standard circle action. No assertion here classifies general Hamiltonian circle actions, reconstructs a symplectic manifold from its moment measure, or extends the sharp arithmetic bound to arbitrary toric polytopes. In particular, the Euler characteristic alone cannot give the lower bound in Theorem 1.1.

Three distinctions matter. Rational rank is not the number of visible levels; rational dependence need not cause a binary subset-sum collision; and a collision need not cancel its coefficient. The corank-one theorem resolves all three alternatives in its stated range. For larger relation spaces, the support bound remains sharp, while a classification of every equality configuration is not supplied here.

The main additional questions are to classify equality for intermediate ranks with several independent relations, and to determine which arithmetic constraints survive for general moment polytopes. These questions are outside the proved statements. The present contribution is the sharp rank bound, its collision-sensitive realization as exact differential complexity, and the complete recurrence divisibility statement for this family.

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